

Instructions:

You will work on these exercises during the practical session in the third week of the course. We will cover exercise 1. We encourage you try it before the session.

Exercise 1: Naive Bayes

Naive Bayes is a classification model that makes strong independence assumptions about the features of the data, conditional on the class label. Specifically, it assumes that each feature is independent given the class label. In contrast, a full-covariance Gaussian class-conditional model can represent correlations between all input features.

If correlated features are treated independently, the evidence for a class may be overcounted. However, Naive Bayes is simple to construct because, by ignoring correlations, the class-conditional likelihood $p(\mathbf{x} | \mathcal{C}_k)$ factors into a product of D univariate distributions:

$$p(\mathbf{x} | \mathcal{C}_k) = \prod_{d=1}^D p(x_d | \mathcal{C}_k).$$

Consider a document classification task with K classes $\mathcal{C}_1, \dots, \mathcal{C}_K$. We represent each document using a bag-of-words vector $\mathbf{x}_n \in \{0, 1\}^D$, where $x_{nd} = 1$ if word d appears in document n , and $x_{nd} = 0$ otherwise. The training inputs form a matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$. The targets form a matrix $\mathbf{T} \in \mathbb{R}^{N \times K}$, whose n -th row is a one-hot vector $\mathbf{t}_n^\top = (t_{n1}, \dots, t_{nK})$ with $t_{nk} = 1$ if document n belongs to class $\mathcal{C}_k$.

Assume the class prior probabilities are known and given by

$$p(\mathcal{C}_k) = \pi_k, \quad \sum_{k=1}^K \pi_k = 1.$$

For this exercise, model word presence/absence using a Bernoulli distribution. Thus, conditioned on class $\mathcal{C}_k$, word d is Bernoulli distributed with parameter

$$\theta_{dk} = p(x_d = 1 | \mathcal{C}_k),$$

and

$$p(\mathbf{x} | \mathcal{C}_k, \theta_{1k}, \dots, \theta_{Dk}) = \prod_{d=1}^D \theta_{dk}^{x_d} (1 - \theta_{dk})^{1-x_d}.$$

- Write down the data likelihood $p(\mathbf{T}, \mathbf{X} | \boldsymbol{\Theta})$ without first using the Naive Bayes independence assumption. Then derive the likelihood for the general K -class Naive Bayes classifier, stating where you use the product rule and the Naive Bayes assumption. Write the likelihood in terms of $p(x_d | \mathcal{C}_k)$, without assuming the explicit Bernoulli form.
- Write down the data log-likelihood $\ln p(\mathbf{T}, \mathbf{X} | \boldsymbol{\Theta})$ for the Bernoulli model.
- Solve for the maximum likelihood estimator of θ_{dk} . Explain how the result can be interpreted.
- Write $p(\mathcal{C}_1 | \mathbf{x})$ for the general K -class Naive Bayes classifier.
- Write $p(\mathcal{C}_1 | \mathbf{x})$ for the Bernoulli model.

Exercise 2: LDA and QDA

Exam question

Suppose we have a dataset consisting of data points with two features, with each data point assigned to one of two classes, $\mathcal{C}_1$ or $\mathcal{C}_2$. Suppose that the class-conditional likelihoods are known:

$$p(\mathbf{x} | \mathcal{C}_1) = \mathcal{N}\left(\mathbf{x} \mid \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\right), \quad p(\mathbf{x} | \mathcal{C}_2) = \mathcal{N}\left(\mathbf{x} \mid \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right).$$

- Were we to perform posterior classification using this model, would it fall in the category of Linear Discriminant Analysis (LDA) or Quadratic Discriminant Analysis (QDA)? Why?
- Assume we are told, before seeing any data, that we should expect twice as many data points in $\mathcal{C}_2$ as in $\mathcal{C}_1$. Write out the posteriors $p(\mathcal{C}_1 | \mathbf{x})$ and $p(\mathcal{C}_2 | \mathbf{x})$ given this information. You can leave the evidence term $p(\mathbf{x})$ in your expressions.

Recall that the multivariate Gaussian density is

$$\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^D |\boldsymbol{\Sigma}|}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right].$$

- Come up with a condition of the form $\mathbf{x}^\top \mathbf{x} > a$ defining when we classify $\mathbf{x}$ as $\mathcal{C}_1$ rather than $\mathcal{C}_2$.

Exercise 3: Classification regions

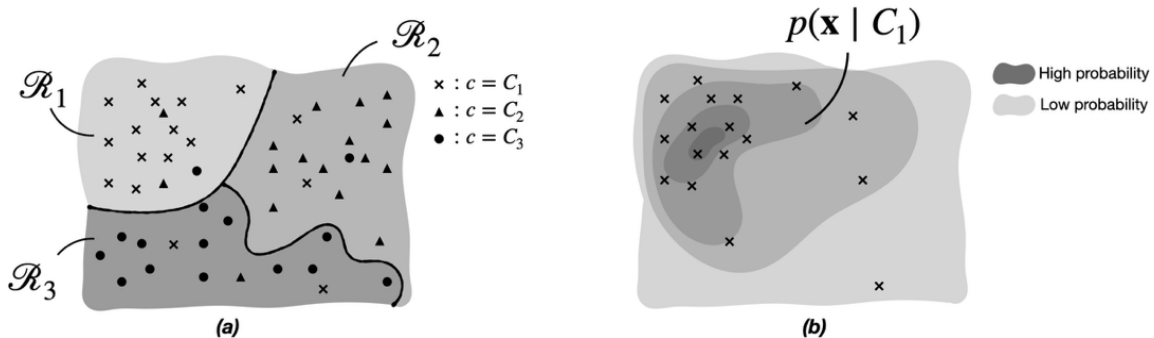
Exam question

Consider a three-class classification problem with points $\mathbf{x} \in \mathbb{R}^2$ that belong to one of three classes $\mathcal{C}_1$, $\mathcal{C}_2$, or $\mathcal{C}_3$. We build a classifier by partitioning the 2D plane into three decision regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$, corresponding to the three classes. That is, we classify a point $\mathbf{x}$ as class $\mathcal{C}_k$ when $\mathbf{x} \in \mathcal{R}_k$.

We assume that the observations follow class-conditional distributions

$$\mathbf{x} \sim p(\mathbf{x} | c = \mathcal{C}_k),$$

which we write more compactly as $p(\mathbf{x} | \mathcal{C}_k)$.



- Suppose we know the class-conditional distribution $p(\mathbf{x} | \mathcal{C}_1)$. Give the probability that a point sampled from this distribution is correctly classified as class $\mathcal{C}_1$.
- Without knowing anything about the class-conditional distributions of the other classes, define the total error rate that the classifier makes on class $\mathcal{C}_1$. Let the total error rate be the fraction of times a point $\mathbf{x}$ is assigned any class other than its true class. Write down the expected error rate for points in class $\mathcal{C}_1$.

- (c) All forms of errors, and the frequencies with which they are expected to occur, can be characterized by an expected confusion matrix. Define the expected confusion matrix in terms of the three class-conditional distributions and the decision regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$.
- (d) For the purpose of classifying points into the three classes, instead of checking in which region $\mathcal{R}_k$ a point falls, one could evaluate the posterior probability that the point belongs to class $\mathcal{C}_k$, assuming that the class priors $p(\mathcal{C}_k)$ are known. Give the expression for $p(\mathcal{C}_k | \mathbf{x})$.
- (e) Suppose we know all true class-conditional distributions. How would you define optimal decision regions? The best possible decision region $\mathcal{R}_i$ for class $\mathcal{C}_i$ is one such that what condition holds?
- (f) Now assume that the data are sampled from the following probabilistic model. Each class-conditional distribution is Gaussian with mean μ_k and covariance matrix Σ_k ,

$$p(\mathbf{x} | \mathcal{C}_k) = \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k),$$

and the prior probability of class $\mathcal{C}_k$ is $p(\mathcal{C}_k) = \pi_k$, where $\sum_{k=1}^3 \pi_k = 1$. Consider a dataset

$$\mathcal{D} = \{(\mathbf{x}_n, \mathcal{C}_n)\}_{n=1}^N,$$

and define the one-hot variable $z_{nk} = 1$ if $\mathcal{C}_n = \mathcal{C}_k$ and $z_{nk} = 0$ otherwise. Write down the likelihood of observing $\mathcal{D}$ under the parameters $\{\mu_k\}$, $\{\Sigma_k\}$, and $\{\pi_k\}$.

- (g) Suppose the optimal decision boundaries are known, as illustrated below. Looking at the decision boundaries, what can we say about the covariance matrices?

